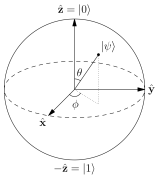
Quantum Computing

$\mathscr{I}$nstructions

- This **2 h** examination consists of 3×4 independent questions of equal weight.
- The use of a non-programmable calculator is permitted, but not especially helpful.
- You may write your answers in French or English (but not both).
- The purpose of this evaluation (just like any other) is to convince us that you are in capacity of saying sensible things on the topics at hand using the appropriate concepts ; please express yourself accordingly !
- Notation : * stands for complex conjugation, † for matrix conjugation and transposition.
- Some useful matrices : $X = \mathrm{NOT} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$, $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$.



Question 1

We apply the transformation $U = \begin{bmatrix} \frac{3i}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{3i}{5} \end{bmatrix}$ to a qubit in the superposed state $|\psi\rangle = \dfrac{|0\rangle + |1\rangle}{\sqrt{2}}$.

a) Explain why we know that this transmormation is *a priori* physically feasible.

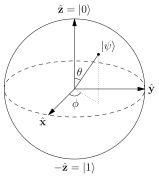
b) Determine the final state $|\psi'\rangle = U|\psi\rangle$ of the qubit et plot these two states on the Bloch sphere.

c) What do we expect to observe if we measure the qubit i) before, and ii) after applying U ?

d) Suppose we have two qubits in the superposed state

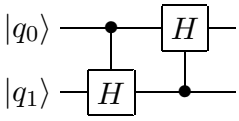
$$|\phi\rangle = \frac{|00\rangle - |01\rangle + 2\,|10\rangle - 2\,|11\rangle}{\sqrt{10}} \in \mathcal{V}_2 \otimes \mathcal{V}_2.$$

Are they entangled ?



Question 2

a) Give the matrix representation for the following quantum circuit :



b) If $f : \{0,1\}^n \to \{0,1\}$ is any (classical) boolean function, recall that U_f denotes the quantum gate on $(n+1)$ qubits for which

$$U_f(|\mathbf{x}\rangle \otimes |z\rangle) = |\mathbf{x}\rangle \otimes |z \oplus f(\mathbf{x})\rangle.$$

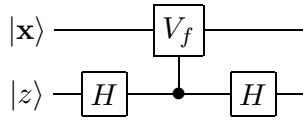
Since it changes the value of $|z\rangle$ according to the value of $f(\mathbf{x})$, U_f is sometimes called a *bit-flip oracle* for f.

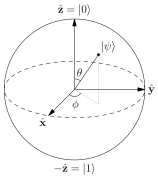
Explicitely give oracles for $f(x,y) = x$ AND y et $g(x,y) = x$ OR y using only elementary quantum gates.

c) For a general function f, compute the effect of U_f on initial states of the shape $|\mathbf{x}\rangle \otimes |+\rangle$ et $|\mathbf{x}\rangle \otimes |-\rangle$, where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad \text{and} \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

d) A *phase-flip oracle* for f is a quantum gate V_f on n qubits for which $V_f |\mathbf{x}\rangle = (-1)^{f(\mathbf{x})}|\mathbf{x}\rangle$. Verify that if we have a phase-flip oracle, the following circuit implements U_f.





Question 3

The Pasqal and D-Wave quantum analog machines take as input Ising or Quadratic Unconstrained Binary Optimization (QUBO) models. As you know, the *Max Cut* problem can be expressed by the objective

$$\min \sum_{(i,j)\in E} Z_i Z_j$$

with E the set of edges and all Z_i taking values in $\{-1,1\}$ for each vertex i in the set X. We now consider another optimization problem : the *Maximum Independent Set* problem (MIS) which can be defined as follow :

"We search the **largest** possible subset of vertices in a given graph, such that **no** two vertices in the subset are connected by an edge."

In Figure 1, the set {1, 4, 6} (Graph C) is one of the optimal solutions of the (MIS).

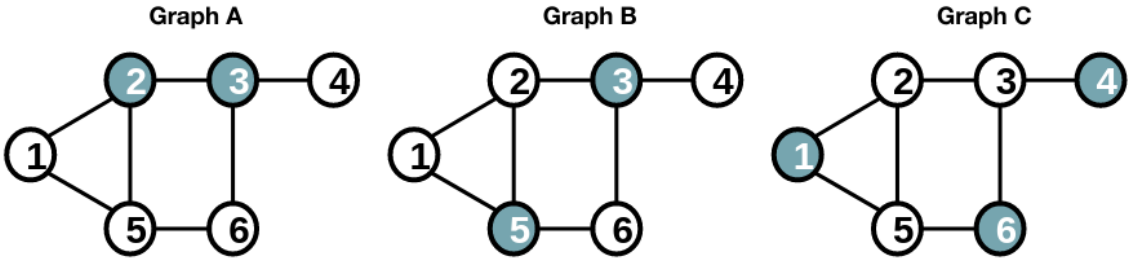


FIGURE 1 – Selected vertices in graphs A, B and C forming a non-feasible solution, a feasible solution, and an optimal solution, respectively.

a) Give all the optimal solutions for the graph in Figure 1.

b) We denote x_i a {0,1} binary variable, with $1 \leqslant i \leqslant |X|$, X the set of vertices, such that $x_i = 1$ if i is in the independent set, $x_i = 0$ otherwise. By using these variables, generalize an objective function maximizing the number of vertices in the independent set.

c) For a given edge connecting two vertices i and j, formulate a quadratic equality that constrains that x_i and x_j cannot be both selected in the independent set. Generalize for all pairs of vertices in the set of edges E.

d) Merge the solutions of questions b) and c) to form a QUBO for the input of an analog machine. Find a mathematical way to avoid non-feasible solutions as much as possible.